Collaborative Incrementally Verifiable Computation



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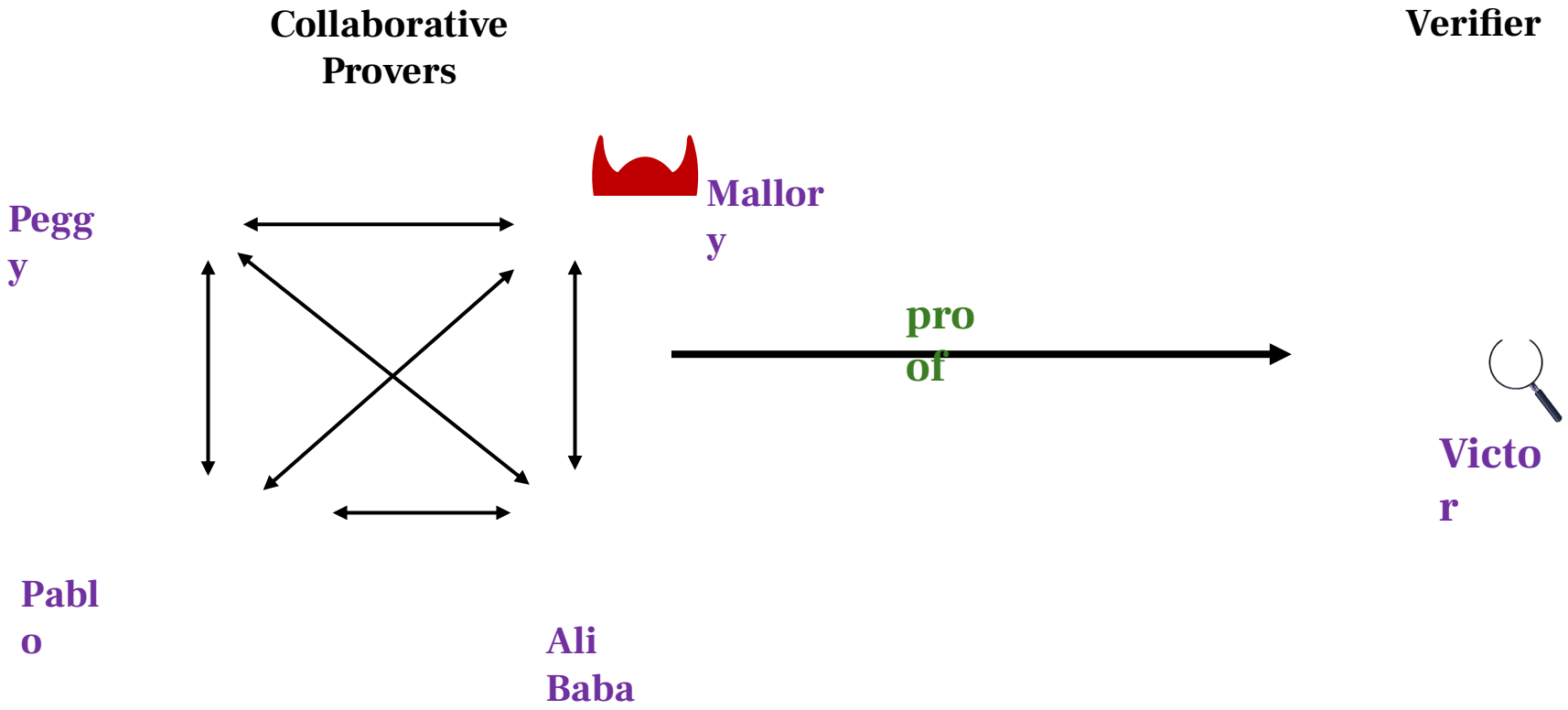
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Stony Brook University



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NTT Research

Collaborative zkSNARKs

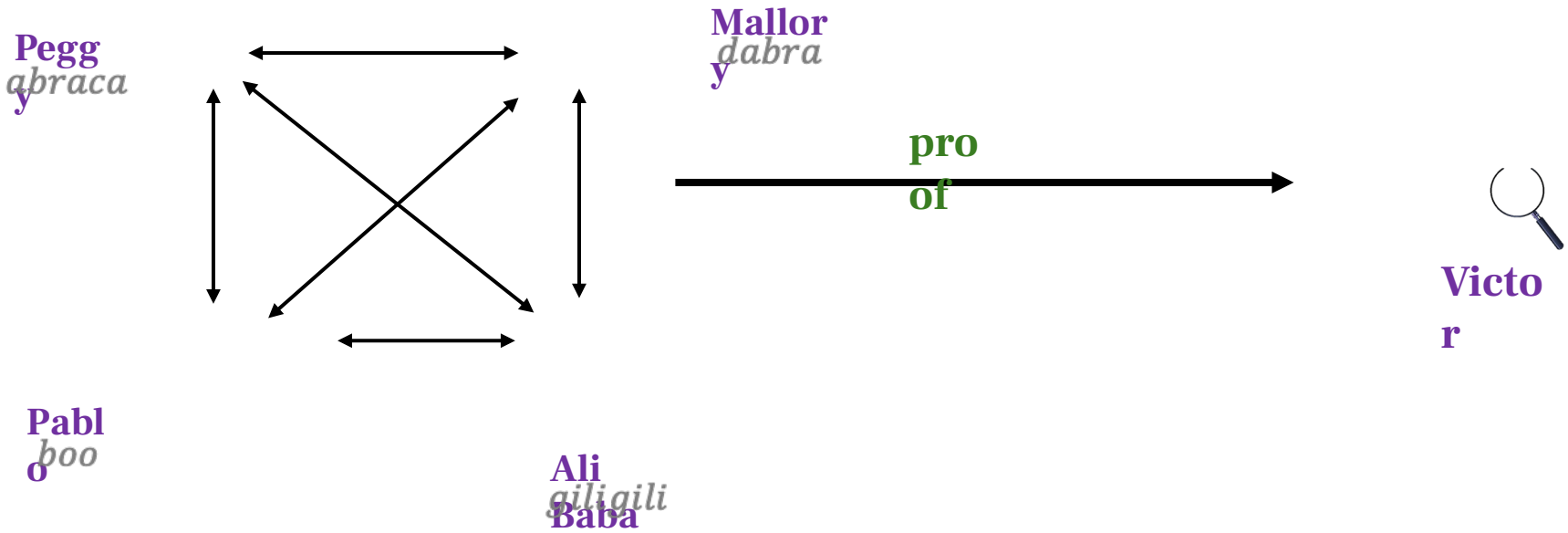
[OB22]



t-Zero-Knowledge: t dishonest provers cannot learn witnesses of other provers

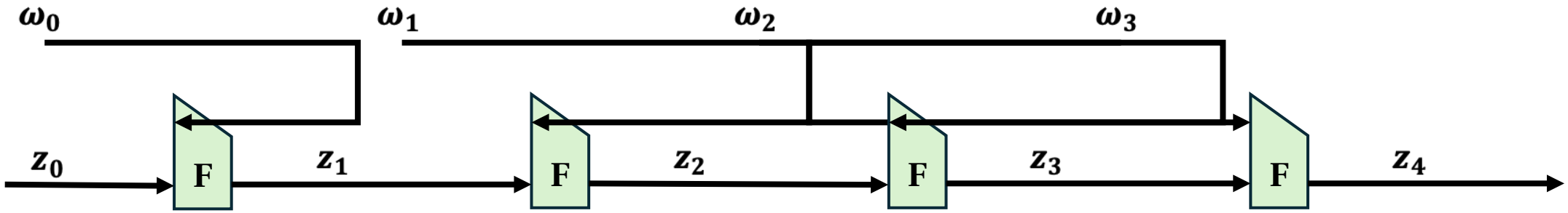
Applications

- Privacy-preserving audits
- Proofs of credit score
- Proofs of training on decentralized datasets



The Streaming Setting

- Apply a simple circuit many times
- Requires small working memory
- Updatable state

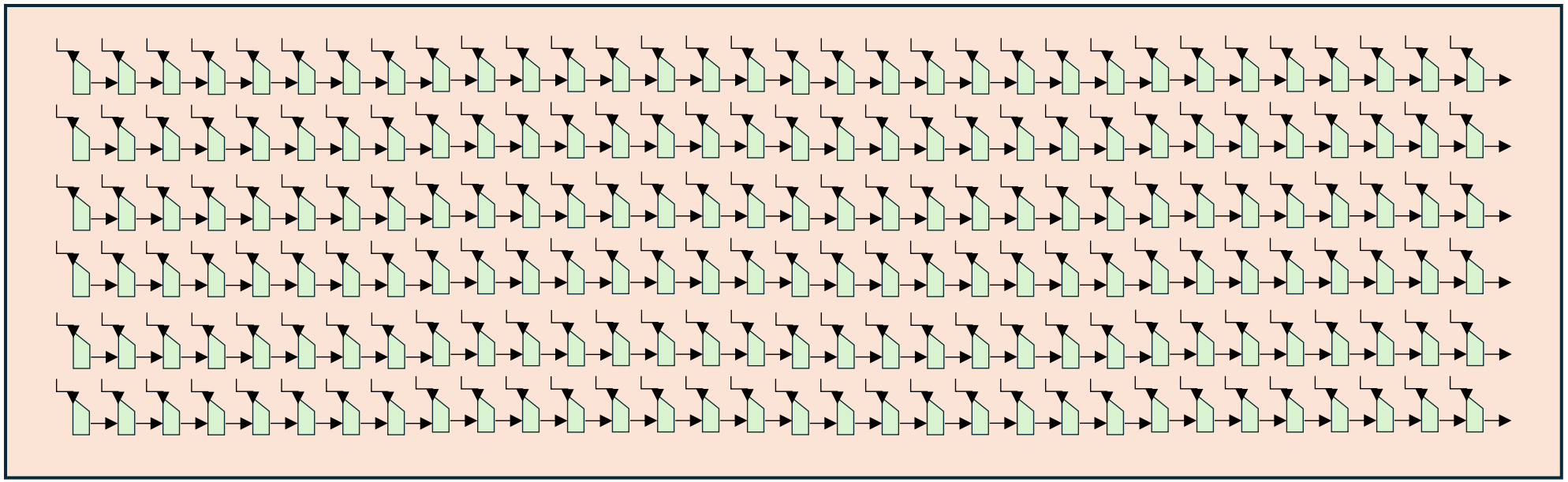


The Streaming Setting

- Small:
  - $|F|$ (size of incremental circuit)
  - $n \approx 5$ (number of provers)

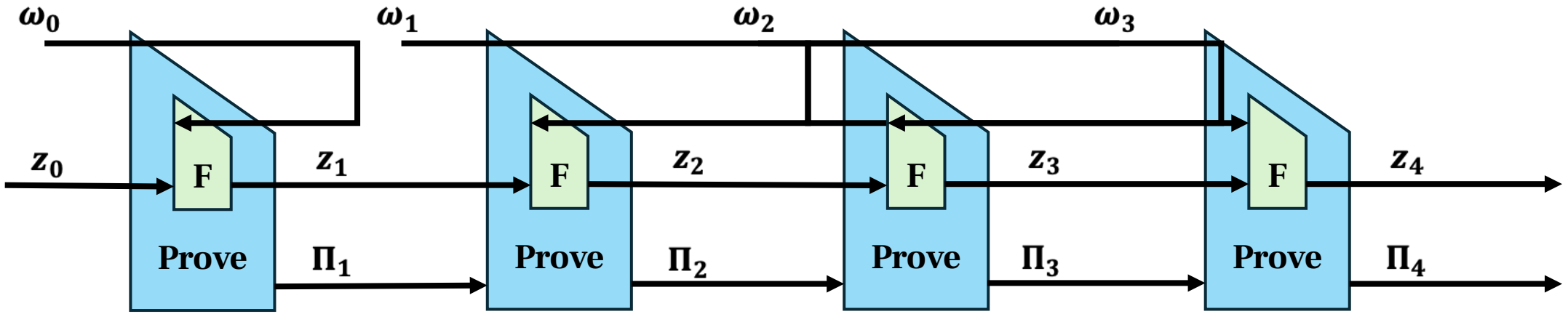
Large:

- k (number of increments)
- $C = k \cdot |F|$ (total circuit size)



Incrementally Verifiable Computation (IVC)

- Optimized for streaming computations
- Updates a running proof
- Working memory grows with $|F|$ not k.



Prior Work

- Provers need large working memory when witness is large
- Proofs are not easily updatable

Scheme	Prover Time	Memory	Per Party Comm.
[OB22]	$O(C \log C)$	$O(C \log C)$	$O(C)$
[LZW+24]	$O(\frac{C}{n})$	$O(\frac{C}{n})$	$O(\frac{C}{n})$
DFS [HMW+25]	$O(C)$	$O(C)$	$O(\log(C))$
[FGR+25]	$O(\frac{C}{n} \log \frac{C}{n})$	$O(\frac{C}{n} \log \frac{C}{n})$	$O(\frac{C}{n})$

Per Party Communication:

- Number of field/group elements broadcast by each party
- Does not include the cost of secret-sharing the witness

k = number of increments

C = Total circuit size

n = number of provers

Our Work

- We construct collaborative IVC in the honest majority setting
- Memory scales with circuit size of **one** increment
- Per-party communication is sublinear in circuit size C
- Malicious security comes for free

Scheme	Prover Time	Memory	Per Party Comm.
[OB22]	$O(C \log C)$	$O(C \log C)$	$O(C)$
[LZW+24]	$O(\frac{C}{n})$	$O(\frac{C}{n})$	$O(\frac{C}{n})$
DFS [HMW+25]	$O(C)$	$O(C)$	$O(\log(C))$
[FGR+25]	$O(\frac{C}{n} \log \frac{C}{n})$	$O(\frac{C}{n} \log \frac{C}{n})$	$O(\frac{C}{n})$
This work	$O(C + kn)$	$O(\frac{C}{k} + n)$	$O(k + \frac{C}{k} + n)$

Per Party Communication:

- Number of field/group elements broadcast by each party
- Does not include the cost of secret-sharing the witness

k = number of increments

C = Total circuit size

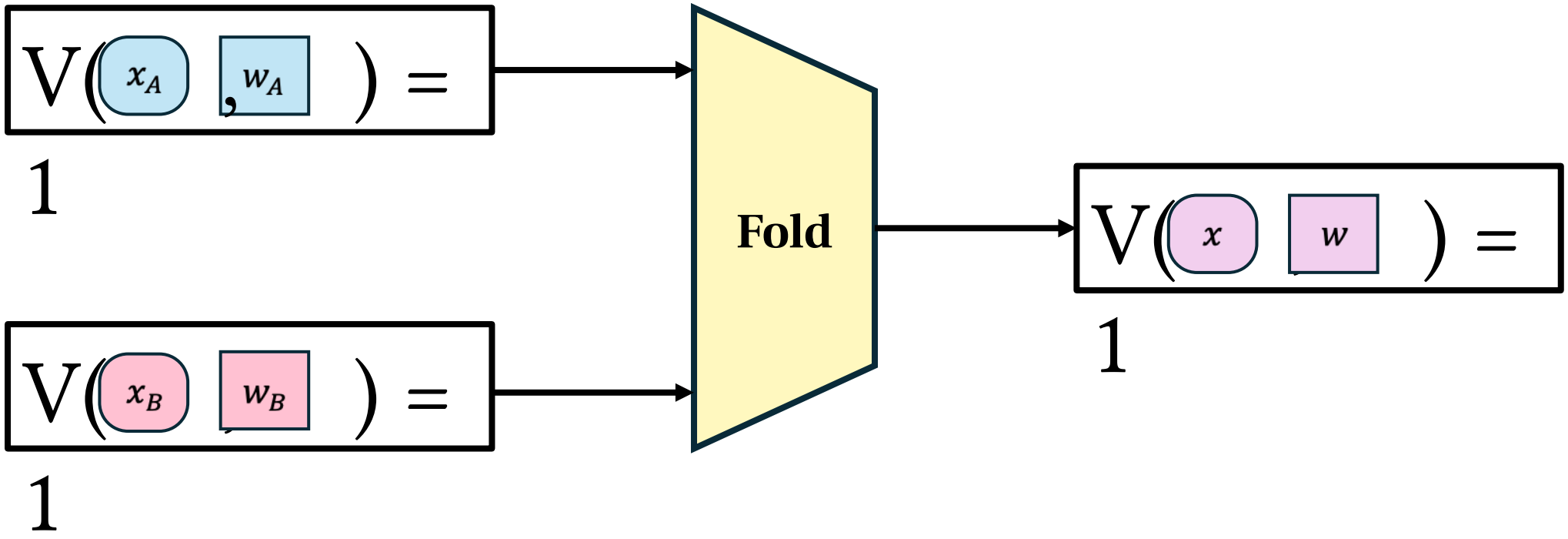
n = number of provers

Folding with Nova

[KST21] - Nova: Recursive Zero-Knowledge Arguments from Folding Schemes

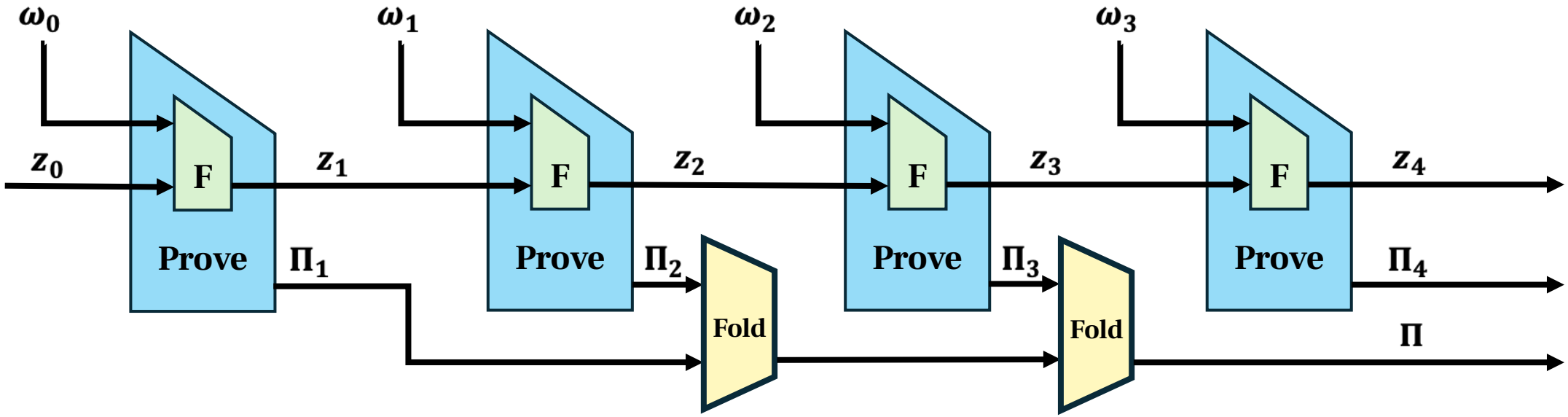
Folding

- Compresses two instance-witness pairs into one



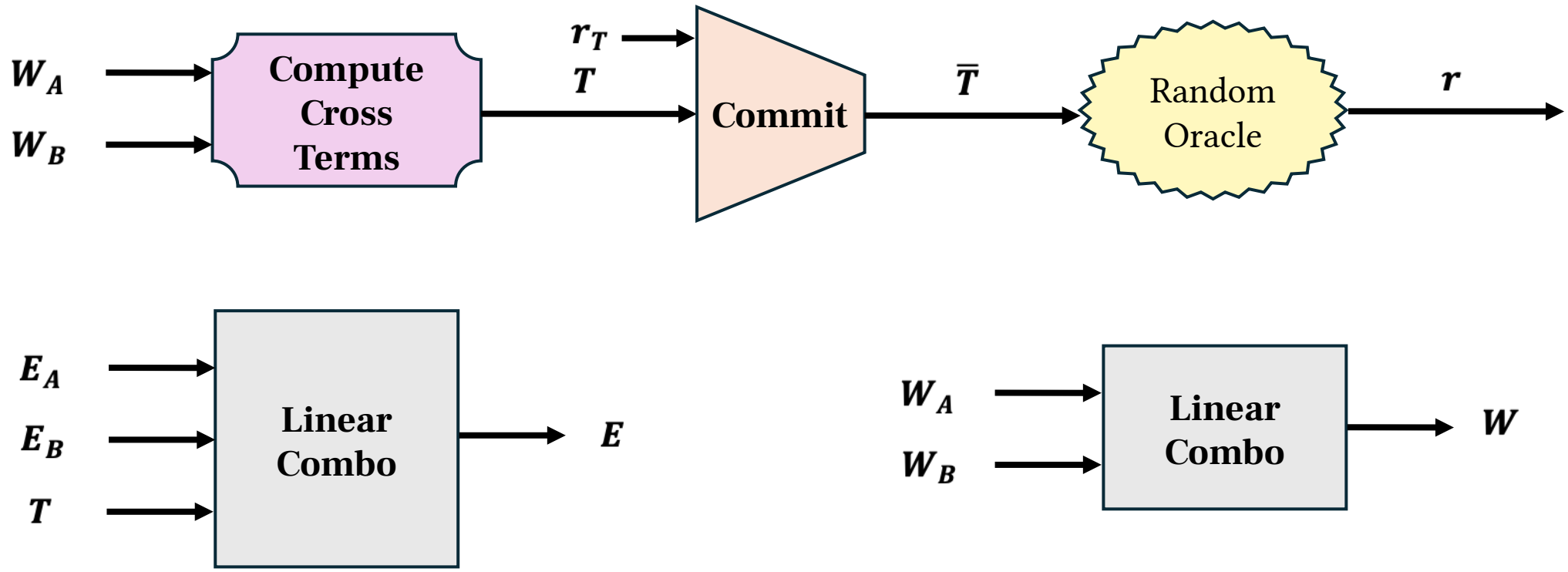
IVC From Folding

- Prove correctness of each increment
- Fold intermediate proofs together



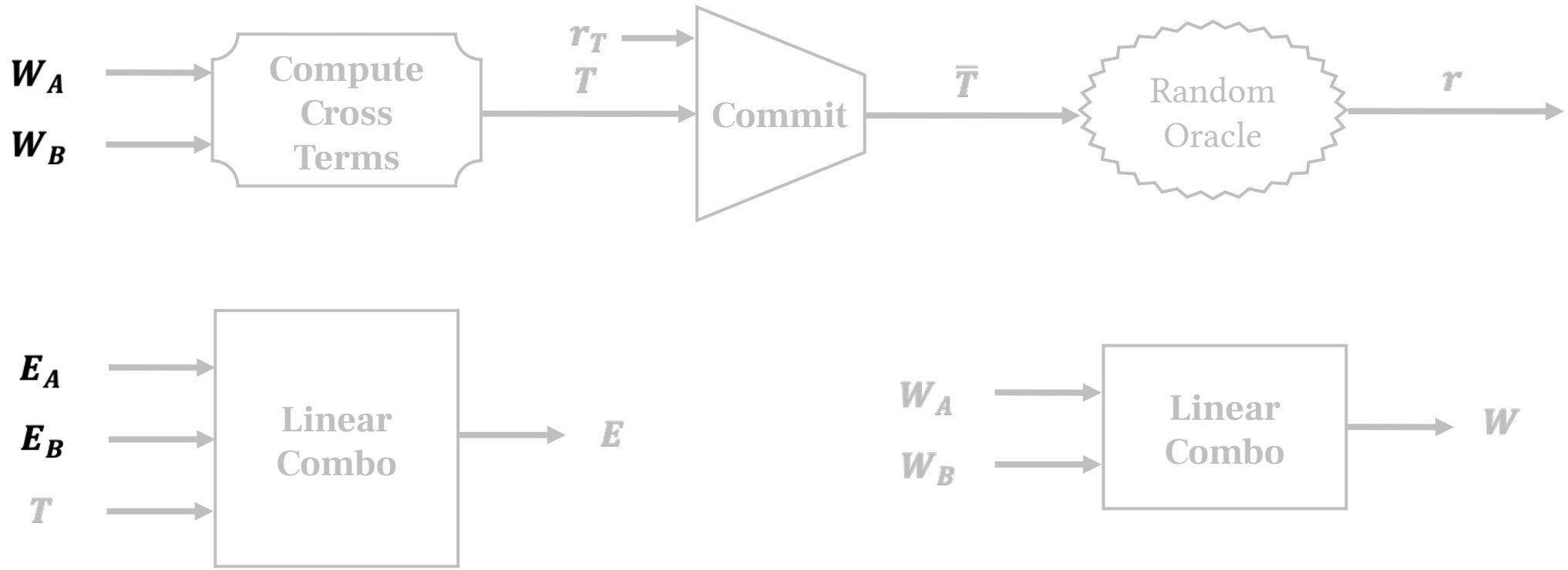
*This picture omits some details from [KST21]'s actual construction

How to Fold


W_A
W_B
Compute Cross Terms
r_T
T
Commit
$\overline{T}$
Random Oracle
r
E_A
E_B
T
Linear Combo
E
W_A
W_B
Linear Combo
W

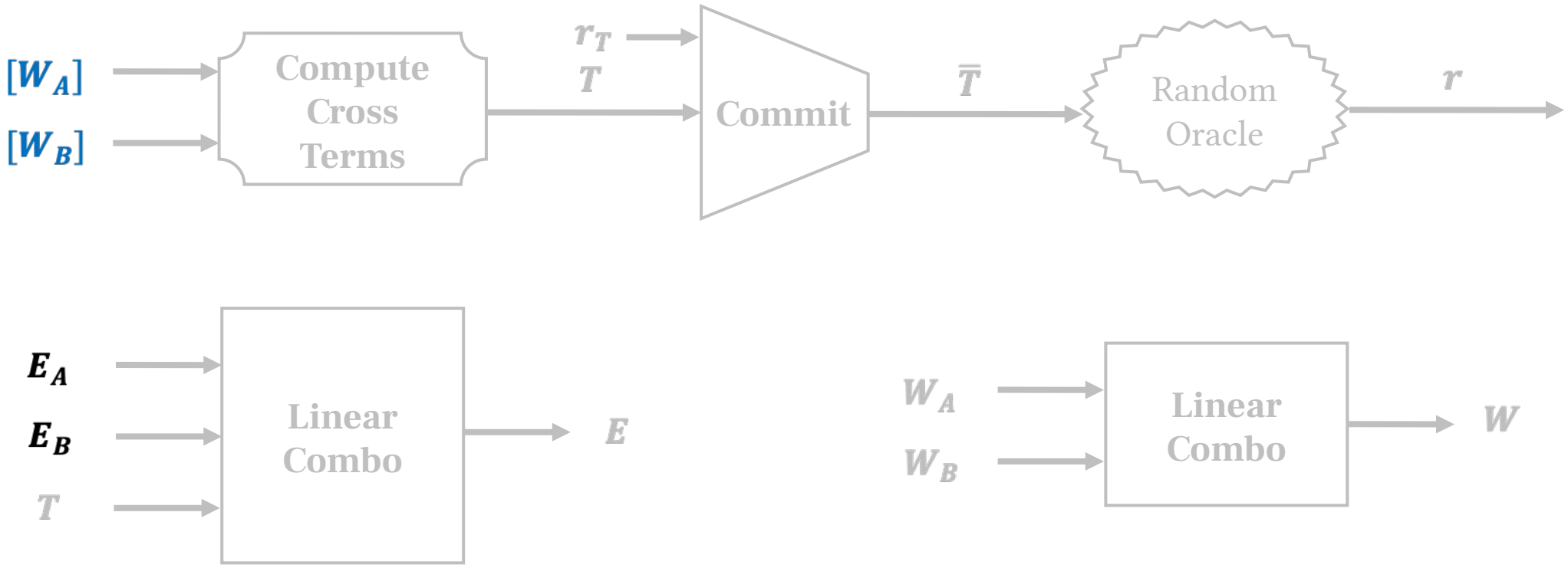
Collaborative Folding

Secret Sharing



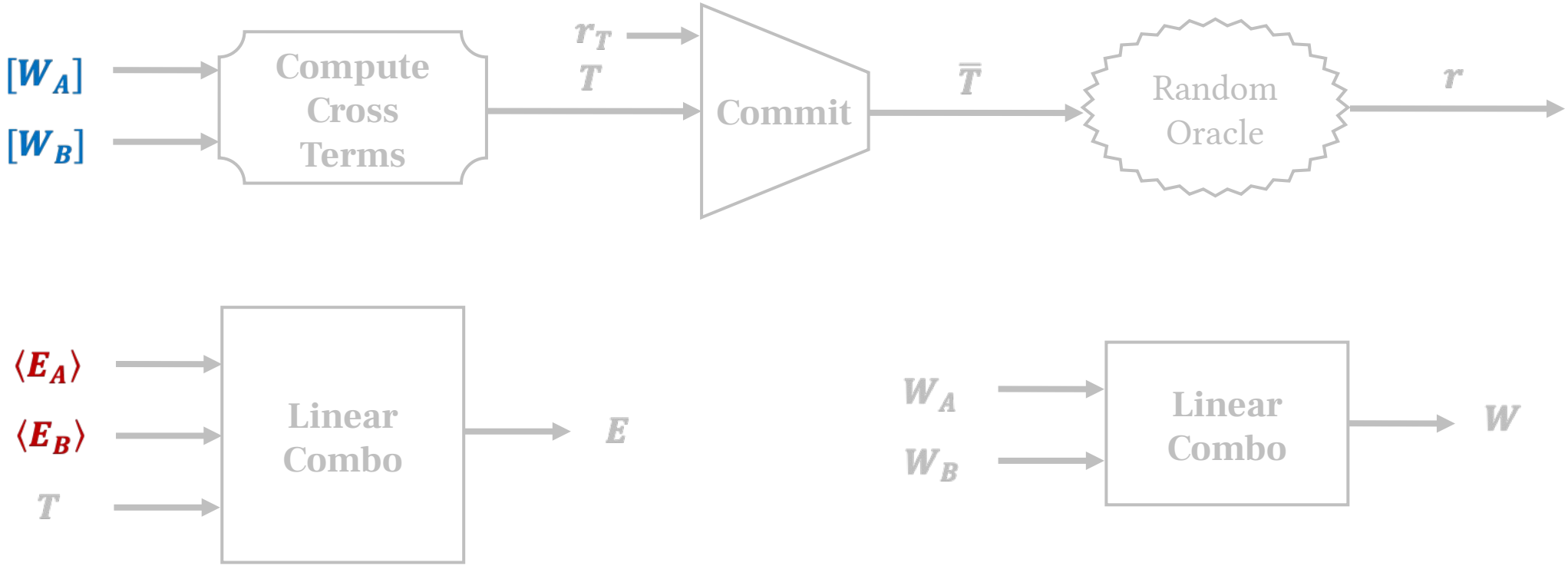
Secret Sharing

- Secret-share W with degree t polynomial



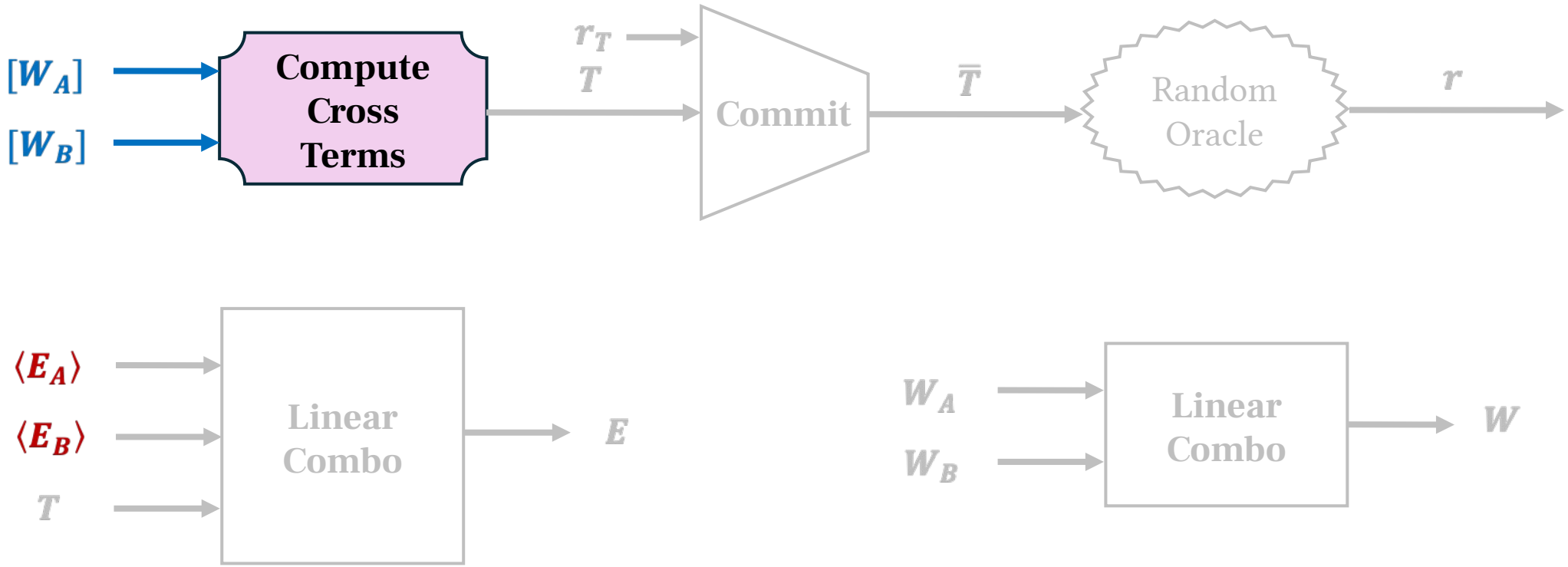
Secret Sharing

- Secret-share W with degree t polynomial
- Secret-share E with degree 2t polynomial



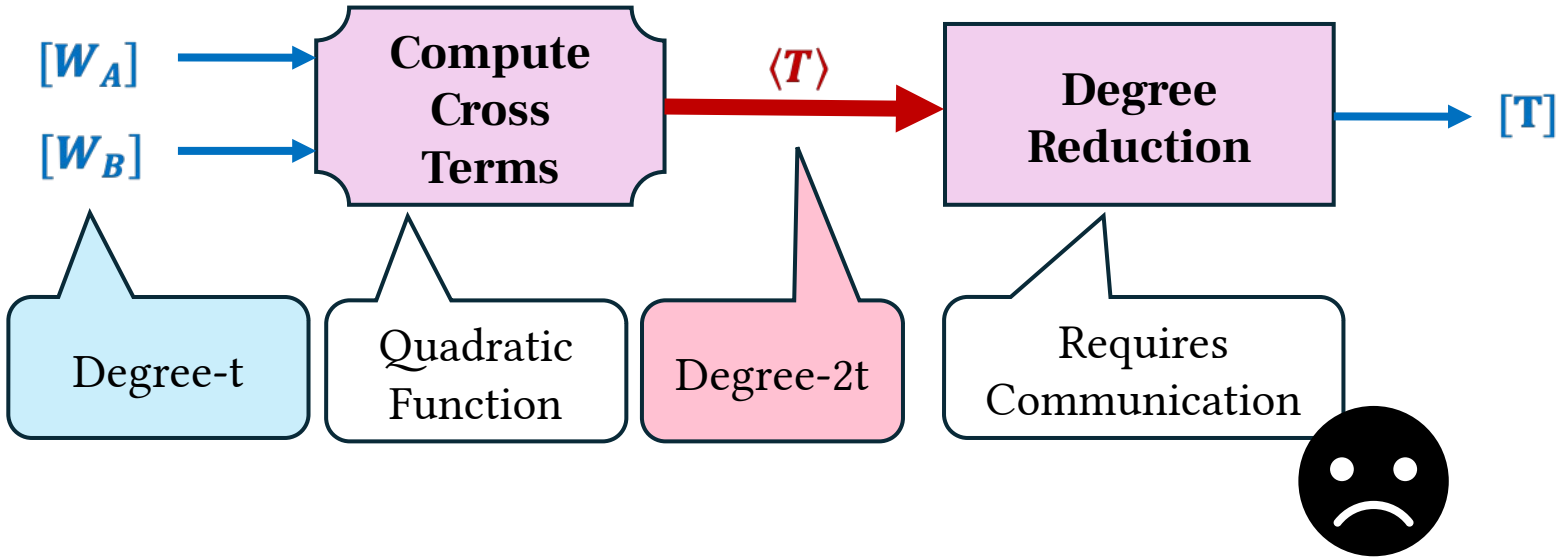
Compute Cross Terms

- Compute the cross terms on secret shared inputs



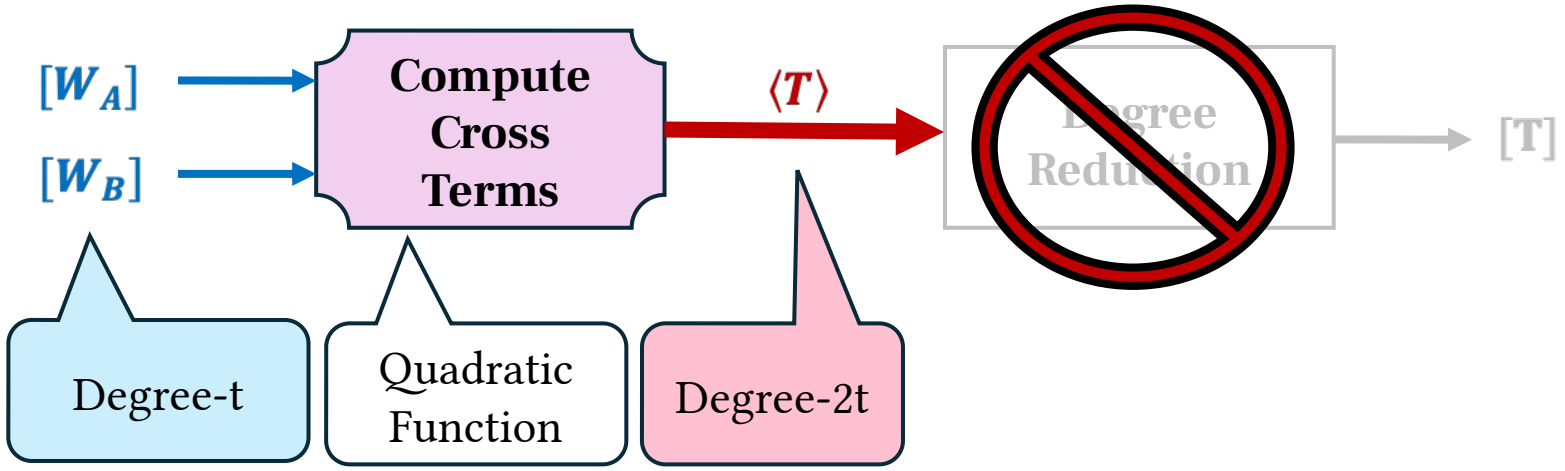
Compute Cross Terms

- Compute the cross terms on secret shared inputs



Compute Cross Terms

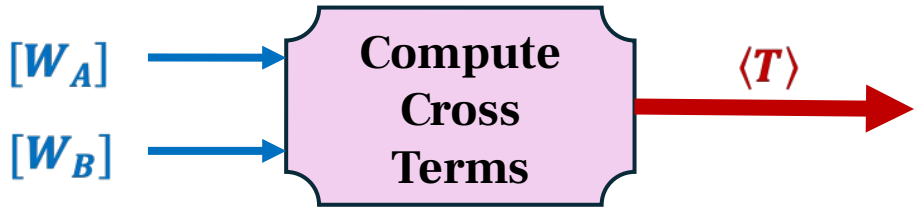
- Compute the cross terms on secret shared inputs
- Don't do degree reduction

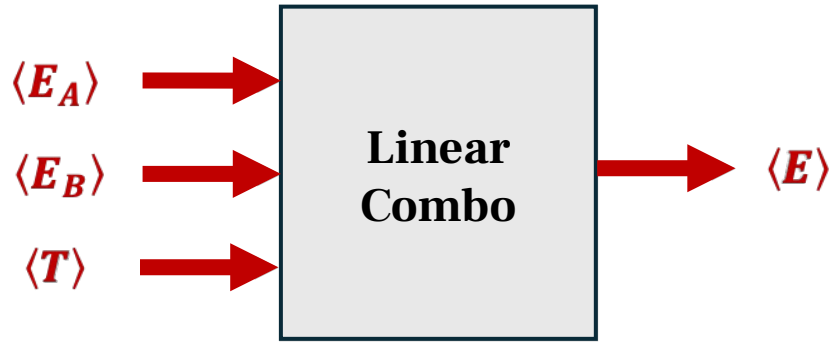




Linear Combination

- Degree doesn't grow during linear combination

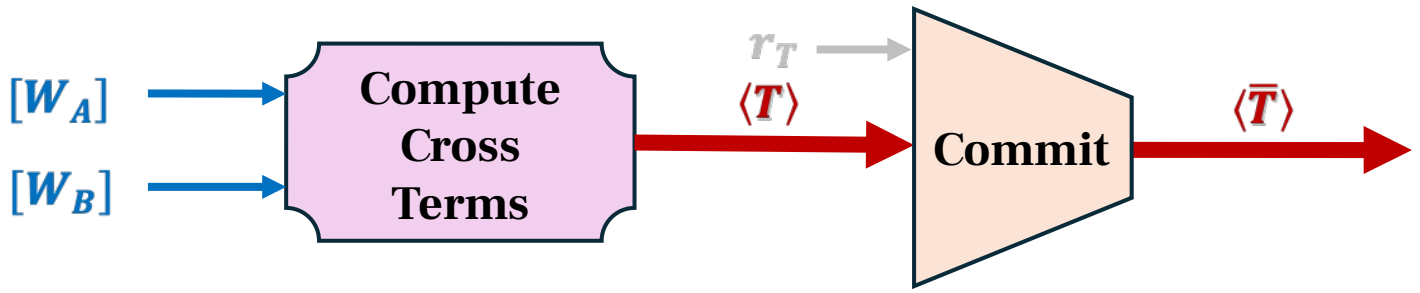






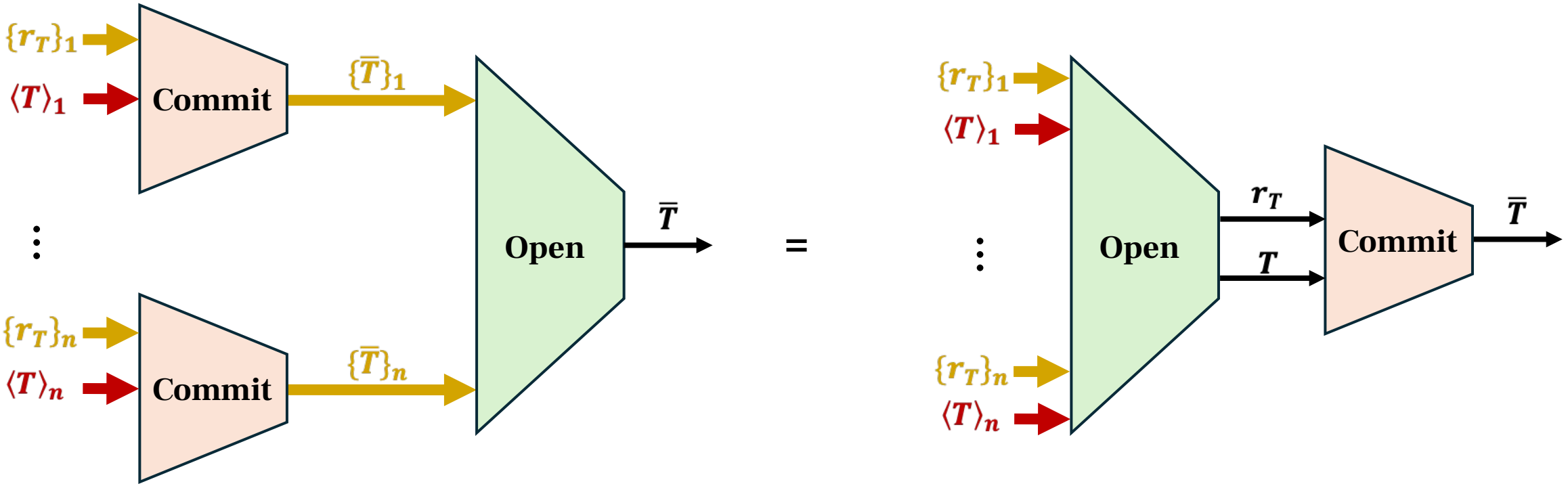
Commit to Cross Terms

- Commit to $\langle \boldsymbol{T} \rangle$

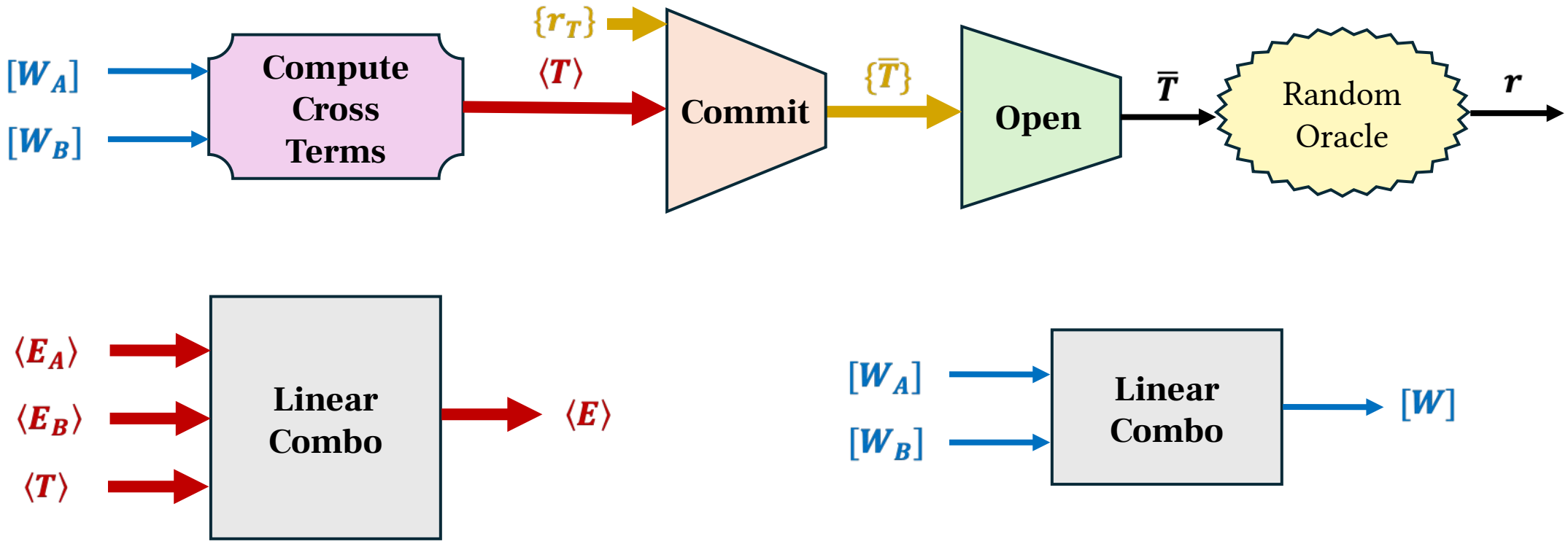


Commit to Cross Terms

- Do MPC in the exponent. Each party commits to their share of T.
- **Commit** is linearly homomorphic, so it commutes with **Open**.

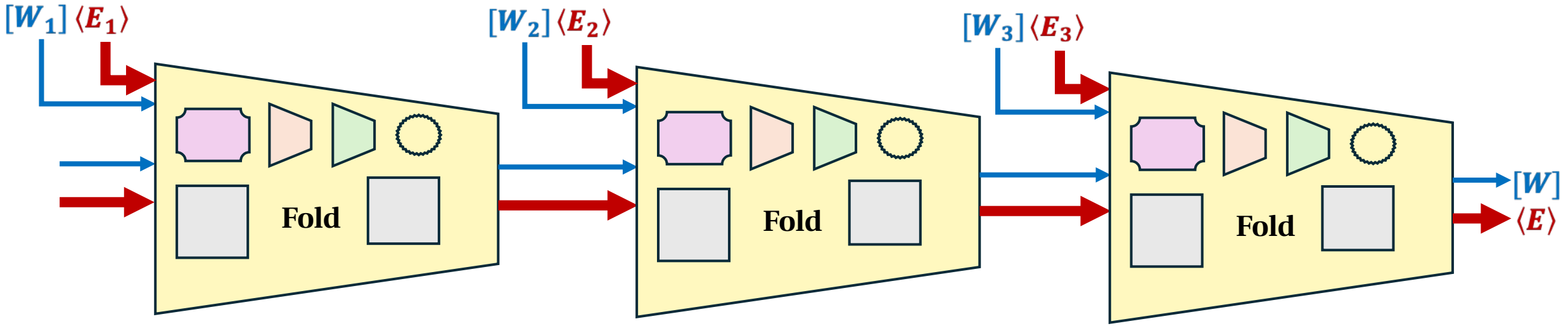


Collaborative Folding



Collaborative Folding

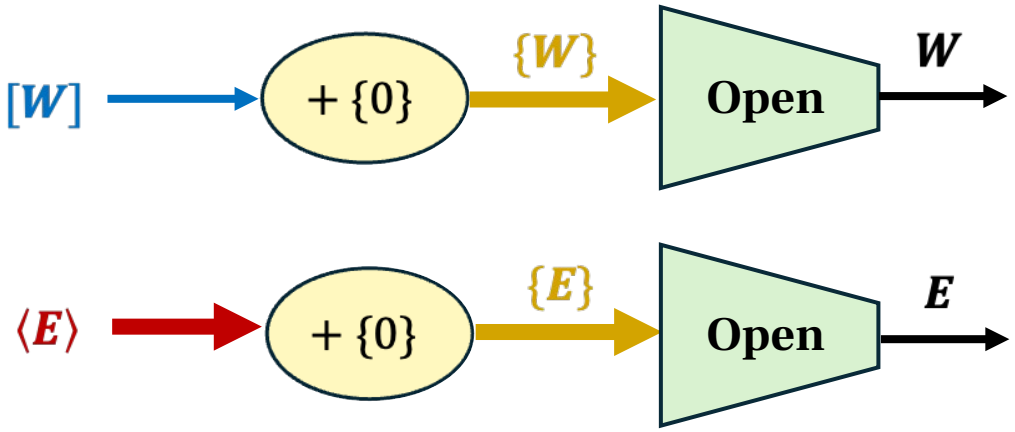
- Perform many rounds of folding on secret-shared values



- Each party broadcasts one group element per round

Reconstruct

- Rerandomize shares of the output
- Then reconstruct



- One-time cost of communication and correlated randomness

Summary

- We construct collaborative IVC in the honest majority setting

Prover Time	Memory	Per Party Comm.
$O(C + kn)$	$O(\frac{C}{k} + n)$	$O(k + \frac{C}{k} + n)$

k = number of increments

C = Total circuit size

n = number of provers

- Malicious security comes for free